Create Project Response #2

1a. **TITLE:** Tristen Erpelding, CSCI 101 Section H, HTML Modular Web Scrapper

1b. **PROGRAM:** The program I am going to write is an HTML web scrapper that will parse the HTML code on a webpage. The user will input a webpage they want to parse. Then the program will print out all the different tags that the user can select from that they want to copy. After the user selects the tags they want, the program will copy the text within the tags and save that data in a file.

```
# URL = 'https://www.geeksforgeeks.org/'
URL = input("Enter URL> ")
# class list set
class_list = set()
# Page content from Website URL
page = requests.get( URL )
# parse html content
soup = BeautifulSoup( page.content,"html.parser" )
# get all tags
# tags = {tag.name for tag in soup.find_all()}
getTags(soup)
```

2a. **COLLECTION:** The program asks the user to enter a URL they want to be parsed of all the HTML. When the user enters the URL, the page is then requested through a GET request which stores that data as text. After that the text file is parsed with the default HTML format. From there a function is called to search through a pseudo text file with all the html in the file getting looked at to find the HTML tags that are present in the HTML.

```
def getTags(webpage):
    allTags = [tag.name for tag in webpage.find_all()]
    resultTags = []
    selectedNum = []
    selectedTags = []

    for item in allTags:
        if item not in resultTags:
            resultTags.append(item)

    print("Select which tags you would like to get data from?")

    for i ,j in enumerate(resultTags):
        if i not in selectedNum:
            print(i,j)

    chocieNum = input("CHOICE> ")
    if chocieNum.isdigit():
        selectedNum.append(int(chocieNum))

    while chocieNum.isdigit():
        for i ,j in enumerate(resultTags):
            if i not in selectedNum:
                print(i,j)
            else:
                selectedTags.append(j)

        chocieNum = input("Choice: ")
        if chocieNum.isdigit():
            selectedNum.append(int(chocieNum))
        else:
            break
    selectedTags = [*set(selectedTags)]
    print(f"The tags you have selected are: {selectedTags}")
    return selectedTags
```

```
Choice: 0
1 meta
2 link
3 script
4 title
6 body
7 div
8 a
9 img
10 span
11 ul
25 path
26 line
27 br
28 u
Choice:
The tags you have selected are: ['style', 'html', 'h2']
```

2b. **ALGORITHM:** The function takes in the webpage and pulls all the tags from the webpage. After all the tags are pulled it removes any repeat tags because when the user is selecting tags, they only need to see it once. The user then chooses the corresponding number to the tag they wish to select. After they have chosen the number, the corresponding tags gets added to a list of selected tags. The loop

then repeats with the new list of tags that can be chosen with the previously selected tag not in the list. When the user is done selecting tags the function prints the list of selected tags and returns the tags to be used later in the program.

Word Count: 177